Maximax and Maximin

Grade 10 – Term 1 – Independent Practice

In every problem, exactly one alternative is chosen before the external state is known. Probabilities are unavailable. All payoffs are profits in thousand USD; negative values represent losses.

1. **Boat Tours.** A boat operator chooses open-sea tours or harbor tours before next month's weather is known. The month will be calm or stormy.

Payoff table (profits in thousand USD)

Alternative	Calm	Stormy
Open-sea tours	46	9
Harbor tours	31	23

Determine the alternative selected by Maximax and by Maximin. Show your calculations.

2. **Translation Agency.** A translation agency chooses to bid abroad or serve local clients before a foreign tender round. Tender access will be open or restricted.

Payoff table (profits in thousand USD)

Alternative	Open	Restricted
Bid abroad	58	7
Local clients	38	24

Determine the alternative selected by Maximax and by Maximin. Show your calculations.

3. **Film Rights.** A film distributor chooses cinema rights or streaming rights before critics review a film. Reviews will be favorable or unfavorable.

Payoff table (profits in thousand USD)

Alternative	Favorable	Unfavorable
Cinema rights	51	4
Streaming rights	33	19

Determine the alternative selected by Maximax and by Maximin. Show your calculations.

4. **Orchard.** An orchard owner commits the land to one use before seasonal rainfall is known: wet, normal, or dry. A fixed lease guarantees the same net profit in every state because the tenant bears all rainfall risk.

Payoff table (profits in thousand USD)

	Rainfall		
Alternative	Wet	Normal	Dry
Plant berries	84	37	-16
Plant apples	63	46	21
Plant olives	43	34	27
Fixed lease	17	17	17

Determine the alternative selected by Maximax and by Maximin. Show your calculations.

5. **Canal Freight.** A freight operator chooses one plan before canal water levels are known: full, limited, or closed to navigation. A fixed charter pays the same net profit in every state because the client pays all operating and rerouting costs.

Payoff table (profits in thousand USD)

	Canal level		
Alternative	Full	Limited	Closed
Deep barges	94	33	-19
Shallow barges	68	47	23
Road freight	49	41	31
Fixed charter	24	24	24

Determine the alternative selected by Maximax and by Maximin. Show your calculations.

6. **Theatre.** A theatre chooses one use of its hall before a rival festival announces its dates. Date overlap will be none, partial, or full. A studio lease guarantees the same net profit regardless of overlap because the tenant bears attendance risk.

Payoff table (profits in thousand USD)

	Date overlap		
Alternative	None	Partial	Full
Stage musical	89	34	-12
Stage comedy	66	51	20
Host workshops	47	39	33
Studio lease	26	26	26

Determine the alternative selected by Maximax and by Maximin. Show your calculations.

7. **Recycling.** A recycling firm chooses one use of its site before scrap prices are known: high, steady, or low. A storage lease guarantees the same net profit under all scrap prices because its rent and costs are fixed.

Payoff table (profits in thousand USD)

	Scrap prices		
Alternative	High	Steady	Low
Process metals	98	31	-21
Sort plastics	73	53	19
Collect paper	52	43	35
Storage lease	29	29	29

Determine the alternative selected by Maximax and by Maximin. Show your calculations.